

CODE SPORT



In this month's column, we discuss entity disambiguation, which is an important problem in natural language processing.

Welcome to the New Year 2022. It seems like 2021 went by in a whirl with the pandemic cloud hanging over all of us. Hopefully, 2022 will be better as we learn to cope with the changes brought on by the pandemic. We had discussed a number of computer science interview questions in the last column. In this column, we take a break from interview questions and instead focus on a popular natural language processing (NLP) topic, namely, entity disambiguation.

One of the questions we discussed in last month's column was on entity disambiguation. The latter is important for relation extraction. Entity definitions are unambiguous entity concepts typically defined in a structured knowledge store such as a knowledge base or knowledge graph. There are a number of structured knowledge bases such as Freebase, YAGO or DBPedia. These knowledge bases contain relations between entities typically expressed as a triple <Barack Obama, President Of, United States> with the two involved entities being <Barack Obama> and <United States>, and the relation between them being <President of>. While knowledge bases like Freebase can provide structured knowledge representations, the amount of knowledge represented in them is limited and static. Hence, AI/NLP systems often look at extracting knowledge from unstructured text in addition to these structured knowledge sources. Wikipedia is a common source of unstructured knowledge, and is often mined for extracting relations from it automatically to populate structured knowledge sources.

Relation extraction is often a challenging task. Unstructured text contains entity mentions that need to be identified as entities first and then disambiguated to match the mention to an entity definition. Consider the examples given below.

Ex1: President Obama met French President Macron at the White House today.

Ex2: The White House opposed the complete waiving of student debt.

Assume that the knowledge base contains an entity definition 'White House' with the definition being: "The physical location or building where the American president resides." In the first sentence, we have an entity mention of White House which matches with this definition. Hence, this entity mention can be disambiguated to the entity definition of the White House as a physical location. The second sentence also contains an entity mention 'White House'. However, this entity mention does not map to the physical location of the 'White House' but refers to the American president who occupies the White House. Hence, this entity mention cannot be disambiguated to the entity definition of White House as a physical location, even though the surface form of both the entity mentions match the entity definition.

We also have the case wherein the surface form of the entity mentions can be completely different from the entity definition due to inter-sentence references or context. For example, consider the sentence: *Potus is considering a ban on travel from high risk Covid countries.* This sentence contains an entity mention 'Potus', which actually refers to the president of the United States though there is no common surface form. Surface forms can completely differ for entity mentions from entity definitions due to inter-sentence references or external knowledge also. Consider the example given below.

Barack Obama is the first person of African-American origin to occupy the White House. The president was born in Hawaii.

In this case, the president is the entity mention which needs to be mapped to the entity definition 'Barack Obama', even though the entity mention surface form has no surface level similarity to the entity definition surface form. Such issues make entity disambiguation a challenging problem.